Markov Chain and Hidden Markov Models

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What Is a Markov Chain (and Why It Matters)?

At the heart of many sequence labeling models lies a concept called the **Markov Chain**.

A Markov Chain is a mathematical model where the probability of a current state (like a label) depends only on the previous state, not the entire history. This is called the **Markov property**.

Example:

Imagine tagging the word “Jones” in this sentence: “Meera Paul met Dr. Jones at Red Cross.”

Using a Markov model, the label for “Jones” is decided based on the label before it. If “Dr.” is labeled as a title, the model infers that “Jones” is likely a person name (PER). The model doesn’t care how we got there, it just uses the most recent label to inform the next one.

Enter Hidden Markov Models (HMMs)

Hidden Markov Models (HMMs) build on Markov Chains by assuming there’s a hidden sequence of states (labels) that generate the observed sequence (the words).

In other words:

- You see a sentence like: “Paul met Dr. Jones.”
- The model assumes there’s a sequence of hidden tags--e.g., PER (Person), ORG (Organization), O (outside) -- that generated those words
- It uses training data to learn how likely certain tags are to emit certain words

This idea of learning the joint probability of both the words and their labels is powerful, but it has limitations.

HMMs:

- Are generative models: They model how data is generated
- Rely heavily on the Markov property: The past influences only the present
- Struggle with long-range dependencies (e.g., linking “Dr.” and “Jones” even if several words apart)

Still, HMMs were foundational in Natural Language Processing and are an important stepping stone toward modern models like CRFs and LSTMs.

Why Learn HMMs in 2025?

You might be wondering: “If deep learning models exist, why study HMMs?”

Here’s why:

- HMMs give you mathematical intuition about sequence prediction
- They form the basis for CRFs, which you’ll use next
- They help you understand label dependencies and probabilistic reasoning, which are essential for error analysis
- Understanding HMMs helps you see how predictions are made, and why your CRF model might succeed or fail

Recommended Readings & Tools

- Speech and Language Processing (Jurafsky & Martin)
  - Chapter 17: Sequence Labeling for Parts of Speech and Named Entities
- Blog: “HMMs for Sequence Tagging” – by Michael Collins Clear walkthrough of HMMs in the context of NLP

Reflection Prompt

Think about a time when a word’s meaning changed based on what came before it, like:

“Apple is tasty” vs. “Apple announced new products.”

- How would a machine decide the label for “Apple”?
- What information should it use?
- What limitations might a simple Markov model face?

Assignment Preview

As you move into training HMMs and working with CRFs in your notebook practice, keep in mind **how** these models make decisions, not just **what** they predict.

The more you understand the math and structure behind them, the better you’ll be at debugging and improving your pipeline.

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